

Telexiles: End-to-end Remote Transmission of Fabric Tactile Sensation

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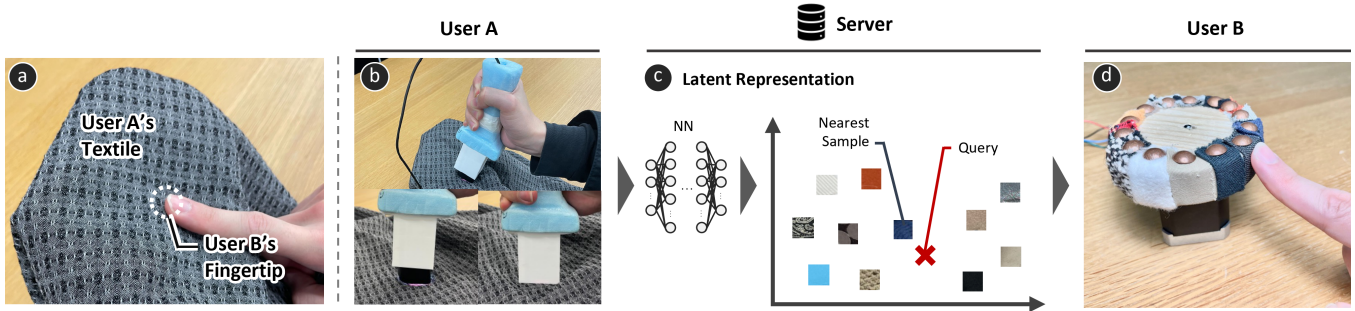


Figure 1: Overview of our Telexiles system. (a) Our ideal of remote tactile transmission. Our Telexiles aims to reproduce the tactile sensation of textiles at a remote location as if they were being touched there. (b) A user transmits the tactile sensation of the textile to the server using tactile sensors. (c) On the server, the latent space is learned in advance from multiple textiles, taking into account their tactile characteristics. The user's data is converted into a latent feature, and the closest textile sample is selected. (d) The other remote user touches a physical sample of the selected textile on the latent space via an actuator.

ABSTRACT

The tactile sensation of textiles is critical in determining the comfort of clothing. In remote scenarios such as online shopping, sensors need to distinguish different garments even with hand-held sensors, and current actuation devices can only present a limited number of known patterns and cannot transmit unknown tactile sensations. We propose Telexiles, an interface for remotely transmitting textile tactile sensations, which uses contrastive self-supervised learning to create a latent space that reflects the relative proximity of textiles. We convert the latent features into a scalar for the one-dimensional structure of the roller. We then select 16 equidistant samples from this line to represent different regions of the latent space. The roller rotates to select the textile with the closest feature. We also show

visually the relationship between a textile touched by a remote user and a set of textiles previously registered in our system.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; *Haptic devices*; *User studies*.

KEYWORDS

Texture recognition; Self supervised learning; Haptic feedback;

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1 INTRODUCTION

The tactile sensation of textiles is critical to an individual's clothing experience. However, when shopping online, customers cannot physically examine the feel of the textiles, making it difficult for

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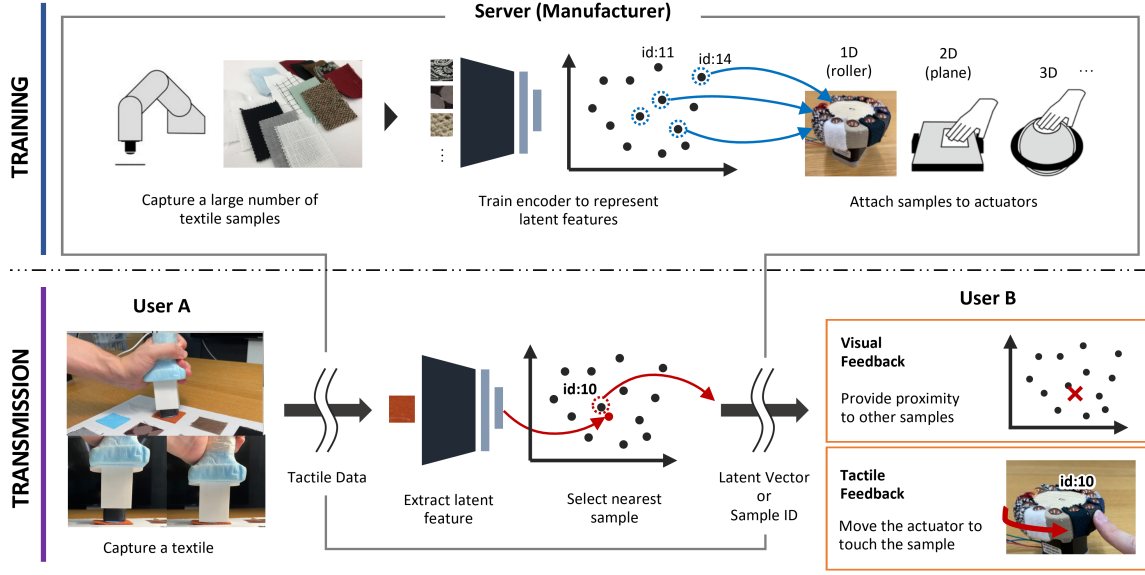


Figure 2: System overview of Telexiles.

them to accurately anticipate the comfort level of the garment. This limitation can reduce the quality of the shopping experience, as the garments purchased may not meet the customer’s comfort expectations. To address the challenges, we seek to develop a system that can transmit the tactile sensation of textiles to a remote location, allowing users to feel as if their fingers have touched the textile on-site.

For remote use, such as online shopping, the sensing device must be able to detect various textiles, even with unstable hand-held sensors. An example of a simple tactile sensor is an elastomer sensor [2] used for the tactile sensing of robot arms. This sensor detects the surface texture of an object by sensing its unevenness when pressed. However, the pressing force and orientation depend on the person, leading to unstable inference. In addition, existing texture recognition methods mainly classify only known textile types [1], making it difficult to determine the relationship between known and unknown textiles that are not in the training data. To reproduce the sensation of textiles, we ultimately want an actuator that can generate all the tactile sensations. However, it is currently difficult to reproduce the tactile sensation of textiles. Haptic Revolver [3] can present a finite number of physical textures on an actuated wheel to physically convey the tactile sensation. However, it does not provide a logic to present the closest tactile sensation when an unknown pattern is an input.

2 TELEXILES SYSTEM

We propose Telexiles, an interface that enables remote transmission of tactile sensations of textiles, even when the input textile is unknown. Our system uses a sensing device that uses contrastive self-supervised learning to create a latent space that reflects the proximity of textiles. We also create an actuation device with 16 textile samples attached according to the proximity obtained by dimensional reduction of the latent features. When a remote user

perceives the texture of the textile with a tactile sensor, the device is actuated to touch the physical sample closest to the user’s fingertips.

Figure 2 shows an overview of our Telexiles system. Our system can be divided into two phases: a training phase, which is performed in advance, and a tactile transmission phase, in which users actually transmit the tactile sensation of the textile. The training phase is assumed to be performed by the manufacturer, who trains encoders to extract a latent vector representation of the tactile sensation from a large sample of textiles. The transmission phase is performed by two remote users, A and B. User A acquires tactile data by pressing a tactile sensor against a textile. This tactile data is sent over the network to the server. Using the learned encoder, the server extracts latent representations from the transmitted tactile data. The server then computes the closest training sample to the transmitted data in latent space and sends the raw latent vector or the ID of the closest sample to User B. User B can confirm the received tactile sensation through visual or tactile feedback. With this sequence of events, Telexile enables end-to-end remote transmission of the tactile sensation of textiles.

3 DISCUSSION AND FUTURE WORK

Our system leverages the power of self-supervised learning to create a continuous latent space representation of material properties, enabling the system to adapt to unknown data and effectively match similar textures. Enhancing the interaction device to support 2D or 3D spatial representations of fabric tactile sensation would enable our system to convey more detailed information about the fabric, further enriching the user experience. These improvements could contribute to the development of more intuitive and user-centered interfaces in a variety of applications.

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